

Symmetric Key Cryptography



UNIT II

SYMMETRIC KEY CRYPTOGRAPHY

MATHEMATICS OF SYMMETRIC KEY CRYPTOGRAPHY: Algebraic structures – Modular arithmetic- Euclid's algorithm- Congruence and matrices -Groups, Rings, Fields- Finite fields- SYMMETRIC KEY CIPHERS: SDES – Block cipher Principles of DES – Strength of DES –Differential and linear cryptanalysis – Block cipher design principles –Block cipher mode of operation – Evaluation criteria for AES – Advanced Encryption Standard – RC4 –Key distribution

long notes

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UNIT II – SYMMETRIC KEY CRYPTOGRAPHY

1. Introduction to Symmetric Key Cryptography

Symmetric key cryptography is a method of encryption where the **same key** is used for both encryption and decryption.

- Sender and receiver share a secret key.
- Encryption converts plaintext into ciphertext.
- Decryption converts ciphertext back into plaintext.

Features

- Fast and efficient
- Suitable for large data encryption
- Requires secure key distribution

Applications

- Secure communication
- Banking systems
- VPN
- Wireless security
- File encryption

2. Mathematics of Symmetric Key Cryptography

Mathematics forms the foundation of cryptography.

2.1 Algebraic Structures

An algebraic structure is a set with one or more operations defined on it.

Examples:

- Addition
- Multiplication
- Modular arithmetic

Important algebraic structures:

1. Groups
 2. Rings
 3. Fields
-

2.2 Modular Arithmetic

Modular arithmetic works with remainders.

Definition

For integers a , b , and n :

$$a \equiv b \pmod{n}$$

if n divides $(a - b)$.

Example

$$17 \pmod{5} = 2$$

because:

$$17 = 5 \times 3 + 2$$

Properties of Modular Arithmetic

Addition

$$(a + b) \pmod{n} = [(a \pmod{n}) + (b \pmod{n})] \pmod{n}$$

Multiplication

$$(a \times b) \mod n = [(a \mod n)(b \mod n)] \mod n$$

Subtraction

$$(a - b) \mod n$$

Importance

- Used in encryption algorithms
- Prevents overflow
- Basis of DES, AES, RSA

2.3 Euclid's Algorithm

Used to find the **Greatest Common Divisor (GCD)**.

Formula

$$\gcd(a, b) = \gcd(b, a \mod b)$$

Continue until remainder becomes 0.

Example

Find GCD of 48 and 18.

$$48 \div 18 = 2 \text{ remainder } 12$$

$$18 \div 12 = 1 \text{ remainder } 6$$

$$12 \div 6 = 2 \text{ remainder } 0$$

Therefore:

$$GCD = 6$$

Importance in Cryptography

- Key generation
- Finding multiplicative inverses

- Used in RSA and finite field arithmetic

2.4 Congruence and Matrices

Congruence

Two numbers are congruent if they leave same remainder.

$$a \equiv b \pmod{n}$$

Example:

$$20 \equiv 2 \pmod{9}$$

because:

$$20 - 2 = 18$$

and 18 is divisible by 9.

Matrices in Cryptography

Matrices are used in:

- Hill cipher
- AES transformations

Example Matrix

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Matrix operations are performed modulo n .

2.5 Groups, Rings and Fields

Group

A group is a set with an operation satisfying:

1. Closure
2. Associativity
3. Identity element

4. Inverse element

Example

Integers under addition.

Ring

A ring has:

- Addition
- Multiplication

Properties:

- Addition forms a group
- Multiplication associative

Example

Integers modulo n .

Field

A field is a ring where:

- Division is possible (except by zero)

Example

Real numbers.

Importance in Cryptography

Fields are used in:

- AES
 - Finite field operations
 - Polynomial arithmetic
-

2.6 Finite Fields

A finite field contains limited number of elements.

Also called:

$$GF(p)$$

(Galois Field)

where p is prime.

Example

$$GF(5) = \{0, 1, 2, 3, 4\}$$

Operations performed modulo 5.

Importance

Used in:

- AES
 - Error correction
 - Modern cryptographic systems
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3. Symmetric Key Ciphers

Two major types:

1. Block Cipher
 2. Stream Cipher
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3.1 S-DES (Simplified DES)

S-DES is a simplified version of DES for educational purposes.

Features

- 10-bit key
 - 8-bit plaintext
 - Produces 8-bit ciphertext
-

Structure

1. Initial permutation
 2. Two rounds
 3. Subkey generation
 4. Final permutation
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Working Steps

1. Generate two subkeys
 2. Apply permutations
 3. Apply Feistel function
 4. Swap halves
 5. Final permutation
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Importance

- Demonstrates DES principles
 - Educational model
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3.2 Block Cipher Principles

A block cipher encrypts fixed-size blocks.

Example:

- DES → 64-bit block
 - AES → 128-bit block
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Principles

Confusion

Makes relationship between key and ciphertext complex.

Diffusion

Spreads plaintext statistics throughout ciphertext.

Introduced by Claude Shannon.

Feistel Structure

Used in DES.

Input divided into:

- Left half
- Right half

Repeated rounds performed.

3.3 DES (Data Encryption Standard)

DES is a symmetric block cipher developed by IBM.

Features

- 64-bit block size
 - 56-bit key
 - 16 rounds
 - Feistel structure
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DES Structure

Steps

1. Initial permutation
 2. 16 rounds
 3. Final permutation
-

Each Round

- Expansion permutation
 - XOR with subkey
 - S-box substitution
 - Permutation
-

Advantages

- Fast in hardware
 - Strong historical algorithm
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Disadvantages

- Small key size
 - Vulnerable to brute-force attack
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3.4 Strength of DES

DES was once highly secure.

Strength Factors

Avalanche Effect

Small plaintext change causes large ciphertext change.

Complex Substitution

S-boxes provide nonlinearity.

16 Rounds

Provides diffusion and confusion.

Weakness

56-bit key too small today.

Brute-force possible using modern computers.

3.5 Differential Cryptanalysis

A cryptanalysis technique studying input differences and output differences.

Idea

- Analyze how plaintext differences affect ciphertext.
- Helps discover secret key.

Process

1. Choose plaintext pairs
 2. Observe ciphertext differences
 3. Analyze probability patterns
-

Impact

DES designed to resist this attack.

3.6 Linear Cryptanalysis

Uses linear approximations to describe cipher behavior.

Principle

Find linear relation among:

- Plaintext bits
 - Ciphertext bits
 - Key bits
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Steps

1. Collect many plaintext-ciphertext pairs
 2. Build linear equations
 3. Guess key bits statistically
-

Impact

Powerful attack against weak ciphers.

3.7 Block Cipher Design Principles

A secure block cipher should satisfy:

Large Key Size

Prevents brute-force attack.

Many Rounds

Improves security.

Strong S-boxes

Provides nonlinearity.

Avalanche Effect

Small change causes large output difference.

Diffusion and Confusion

Essential security properties.

3.8 Block Cipher Modes of Operation

Modes define how block ciphers process long messages.

1. ECB (Electronic Code Book)

Each block encrypted independently.

Advantages

- Simple
- Parallel processing possible

Disadvantages

- Identical plaintext blocks produce same ciphertext.
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2. CBC (Cipher Block Chaining)

Each plaintext block XORed with previous ciphertext.

Features

- More secure than ECB
 - Uses Initialization Vector (IV)
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3. CFB (Cipher Feedback)

Converts block cipher into stream cipher.

Used for:

- Real-time communication
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4. OFB (Output Feedback)

Generates keystream independent of plaintext.

5. CTR (Counter Mode)

Uses counter values for encryption.

Advantages

- High speed
 - Parallelism possible
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3.9 Evaluation Criteria for AES

AES competition considered:

Security

Resistance against attacks.

Performance

Efficient in software and hardware.

Flexibility

Support different platforms.

Simplicity

Easy implementation.

3.10 AES (Advanced Encryption Standard)

AES replaced DES.

Developed from Rijndael algorithm.

Features

- Block size: 128 bits
 - Key sizes:
 - 128-bit
 - 192-bit
 - 256-bit
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Number of Rounds

Key Size	Rounds
128	10
192	12
256	14

AES Structure

1. SubBytes

Byte substitution using S-box.

2. ShiftRows

Rows shifted cyclically.

3. MixColumns

Column mixing operation.

4. AddRoundKey

Round key XORed with state.

Advantages

- Very secure
 - Fast
 - Widely used
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Applications

- Wi-Fi security
 - Banking
 - SSL/TLS
 - Government systems
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3.11 RC4

RC4 is a stream cipher developed by Ron Rivest.

Features

- Variable key length
 - Byte-oriented
 - Fast software implementation
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Working

1. Initialize state array

2. Generate pseudorandom keystream
 3. XOR with plaintext
-

Advantages

- Simple
 - Fast
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Disadvantages

- Security weaknesses
 - Not recommended today
-

Applications

Previously used in:

- SSL
 - WEP
 - WPA
-

3.12 Key Distribution

Key distribution means securely sharing secret keys between sender and receiver.

Methods of Key Distribution

1. Physical Delivery

Key manually delivered.

2. Trusted Third Party

Central authority distributes keys.

3. Public Key Encryption

Symmetric key encrypted using public key cryptography.

4. Diffie-Hellman Key Exchange

Allows secure key sharing over insecure channel.

Challenges

- Secure transfer
 - Key storage
 - Scalability
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Advantages of Symmetric Key Cryptography

1. Fast encryption
 2. Efficient for large data
 3. Less computational overhead
 4. Suitable for real-time systems
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Disadvantages

1. Key distribution problem
 2. Poor scalability
 3. Single key compromise affects security
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Conclusion

Symmetric key cryptography is one of the most important areas of network security. Mathematical concepts like modular arithmetic, groups, rings, fields, and finite fields form the basis of cryptographic algorithms. Modern symmetric algorithms such as DES and AES provide confidentiality and secure communication. AES is currently the most widely used symmetric encryption standard because of its high security and efficiency.